Today

- Position and momentum
- Expectation value
- The uncertainty principle
- Commutation relations
- Time-independent Schrödinger equation

1.11 Position and momentum

Let us focus on a particle in one-dimensional space with coordinate x. Now $|\psi(x,t)|^2$ represents the probability density for finding the particle at point x at time t, and $|\psi(x,t)|^2 dx$ is the probability for finding the particle between x and $x+dx$. Observe that $|\psi(x,t)|^2$ is a real number with unit of $m^{-1/2}$ that can be plotted as a function of x at a given instant of time. Suppose we have a profile of $|\psi(x,t_1)|^2$ shown in the figure below at some time t_1 for a system containing a single particle.

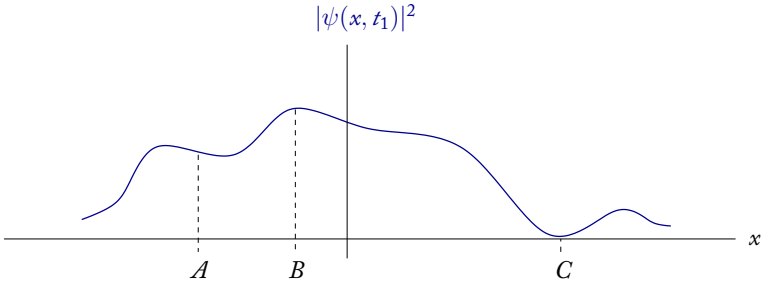


Figure 1.11.1: Probability density at instant of time t_1.

Of all the points on x, the particle is most likely to be found at B, where the probability density is largest, and least likely to be found at C, where the probability density vanishes. The probability for finding the particle anywhere between points A and B at time t_1 is

$$P_{AB}(t_1) = \int_A^B |\psi(x,t_1)|^2 dx, \tag{1.11.1}$$

which represents the area under the curve between the two points. The total probability for finding the particle at any point is $\int_{-\infty}^{\infty} |\psi(x,t_1)|^2 dx = 1$, which is just the total area under the

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curve.

Now we can think about two scenarios. One is preparing a large number of identical systems and measuring the position of the particle for each. The other is repreparing the same system again and again and repeating the experiment of measuring the position of the particle. [Note that this is not the same as measuring the location of the same particle again and again without repreparing the system.] In either case, the average value (also called the expectation value) of the position of the particle is

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \hat{x}\mathcal{P} dx = \int_{-\infty}^{\infty} x|\psi(x,t)|^2 dx \tag{1.11.2}$$

Next we want to write an expectation value, $\langle p \rangle$, for the momentum of the particle. Consider

$$\begin{aligned}
\frac{d}{dt}\langle \hat{x} \rangle &= \frac{d}{dt}\int_{-\infty}^{\infty} x|\psi(x,t)|^2 dx = \int_{-\infty}^{\infty} x\frac{\partial \mathcal{P}}{\partial t} dx = -\int_{-\infty}^{\infty} x\frac{\partial \mathcal{J}}{\partial x} dx \\
&= -\int_{-\infty}^{\infty} \left[\frac{\partial}{\partial x}(xJ) - J\right] dx = -(xJ)|_{x=-\infty}^{\infty} + \int_{-\infty}^{\infty} J dx = \int_{-\infty}^{\infty} J dx \\
&= -\frac{i\hbar}{2m}\int_{-\infty}^{\infty} \left(\psi(x,t)^* \frac{\partial \psi(x,t)}{\partial x} - \psi(x,t)\frac{\partial \psi(x,t)^*}{\partial x}\right) dx \\
&= -\frac{i\hbar}{2m}\int_{-\infty}^{\infty} \left(\psi(x,t)^* \frac{\partial \psi(x,t)}{\partial x} - \frac{\partial}{\partial x}(\psi(x,t)^*\psi(x,t)) + \psi(x,t)^* \frac{\partial \psi(x,t)}{\partial x}\right) dx \\
&= -\frac{i\hbar}{m}\int_{-\infty}^{\infty} \psi(x,t)^* \frac{\partial \psi(x,t)}{\partial x} dx = \frac{1}{m}\int_{-\infty}^{\infty} \psi(x,t)^* \frac{-i\hbar \partial \psi(x,t)}{\partial x} dx.
\end{aligned} \tag{1.11.3}$$

The expectation value of momentum is asserted to be that of the mass of the particle times the rate of change of the expectation value of the position,

$$\langle \hat{p} \rangle \equiv m\frac{d}{dt}\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \psi(x,t)^* \frac{-i\hbar \partial \psi(x,t)}{\partial x} dx = \int_{-\infty}^{\infty} \psi(x,t)^* \hat{p} \psi(x,t) dx, \tag{1.11.4}$$

where, again, $\hat{p} = -i\hbar\frac{\partial}{\partial x}$ is the momentum operator.

Similarly, we can rewrite the expectation value of position as

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \psi(x,t)^* \hat{x} \psi(x,t) dx. \tag{1.11.5}$$

Denoting either one of the position or the momentum operator or any other function of position

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and momentum, and generalizing to three dimensions, by $\hat{O}(\hat{\vec{r}}, \hat{p}) = \hat{O}(\vec{r}, -i\hbar\vec{\nabla})$, the expectation value of $\hat{O}$ is given by

$$\langle \hat{O} \rangle = \int \psi(\vec{r}, t)^* \hat{O}(\vec{r}, -i\hbar\vec{\nabla})\psi(\vec{r}, t) d^3\vec{r}, \tag{1.11.6}$$

where the operator is inserted in-between $\psi(\vec{r}, t)^*$ and $\psi(\vec{r}, t)$

1.12 The uncertainty principle

Let us recall our discussion on double-slit electron diffraction before and after measurement.

Question 1.12.1: The figure below shows $|\psi|^2$ before and after measurement.

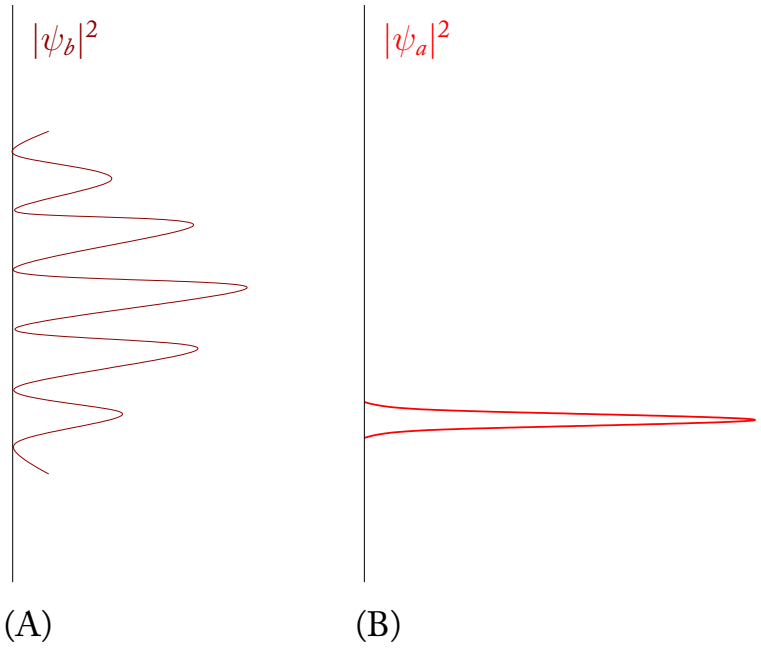


Figure 1.12.1: Probability density before (A) and after (B) measurement.

In which case is the particle's location better determined? What about the wavelength? It turns out, generally, that when the position of the particle is better determined, is momentum is not and vice versa. In order to quantify this uncertainty in measurements of position and momentum, we use the standard deviation for each, Δx and Δp, which is defined as

$$\Delta x = \sqrt{\langle (\hat{x} - \langle \hat{x} \rangle)^2 \rangle}, \qquad \Delta p = \sqrt{\langle (\hat{p} - \langle \hat{p} \rangle)^2 \rangle} \tag{1.12.1}$$

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Using $\hat{O}$ for either one of $\hat{x}$ or $\hat{p}$, we note that

$$(\Delta O)^2 \quad = \quad \langle(\hat{O} - \langle\hat{O}\rangle)^2\rangle = \langle\hat{O}^2 + \langle\hat{O}\rangle^2 - 2\hat{O}\langle\hat{O}\rangle\rangle = \langle\hat{O}^2\rangle - \langle\hat{O}\rangle^2. \tag{1.12.2}$$

The uncertainty principle for position and momentum, which we will investigate further later as the course progresses, states that

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}, \tag{1.12.3}$$

and it quantifies our observation of the uncertainty in position becoming larger when the uncertainty in moments is smaller and vice versa.

To get some physical intuition, let us consider an electron. In order to find its location we have to see it by shining light. The scattering of light from a target has the same as the diffraction from a hole of the same size as the target. [This is called Babinet's principle in optics] Therefore, considering a slit of the size of the electron Δx, we have the diffraction pattern shown in the figure below.

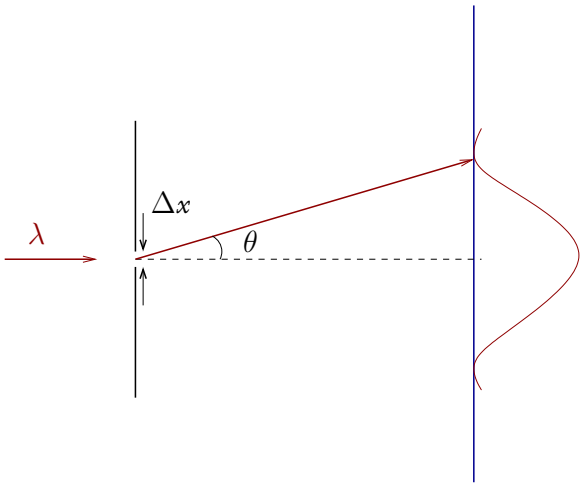


Figure 1.12.2: Scattering of a photon by an electron of size Δx can be viewed as diffraction of a photon by a slit of width Δx.

Because $\sin\theta = \frac{\lambda}{\Delta x}$, θ gets larger as Δx gets smaller. But the component of the magnitude of momentum of the photon along x varies from $p_x = 0$ when it goes straight undeflected to $p_x = \frac{h}{\lambda}\sin\theta$ when it emerges at θ. In other words the range of $\Delta p_x = \frac{h}{\lambda}\sin\theta$ (and the uncertainty in Δp_x) increases with increasing θ, which corresponds to decreasing Δx. Because the total momentum of the photon electron system along x must stay zero, the magnitude of the electron's momentum changes by the same amount. Thus, when Δx is better measured and

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the electron's position is well known, Δp_x is larger the electron's momentum is more uncertain. This is all because of wave-particle duality. In particular, if Planck's constant were to be zero and the wave property of the electron were not there, both its position and momentum could be measured with arbitrary precision. Therefore, $\Delta p_x = \frac{h}{\lambda}\sin\theta = \frac{h}{\lambda}\frac{\lambda}{\Delta x} = \frac{h}{\Delta x}$, or $\Delta x \Delta p_x = h$, consistent with $\Delta x\,\Delta p \geq \frac{\hbar}{2}$ and that we cannot do better than $\Delta x\,\Delta p = \frac{\hbar}{2}$. The uncertainty is intrinsic to quantum mechanics and not of any specific measuring device or scheme.

Question 1.12.2: Based on our investigation so far what is the reason that leads to the uncertainty principle $\Delta x \Delta p \geq \frac{\hbar}{2}$?

1.13 Commutation relations between position and momentum

We have already seen that the position and the momentum operators are given by

$$\hat{\vec{r}} = \vec{r}, \qquad \hat{\vec{p}} = -i\hbar\vec{\nabla}. \tag{1.13.1}$$

Now consider applying $\hat{x}$ first and $\hat{p}_x$ second on the wavefunction $\psi(x,t)$,

$$\hat{p}_x\hat{x}\psi(x,t) = \hat{p}_x\left(\hat{x}\psi(x,t)\right) = -i\hbar\frac{\partial}{\partial x}\left(x\psi(x,t)\right) = -i\hbar\psi(x,t) - i\hbar x\frac{\partial}{\partial x}\psi(x,t). \tag{1.13.2}$$

Next, suppose we reverse the order of operations by applying $\hat{p}_x$ first and $\hat{x}$ second,

$$\hat{x}\hat{p}_x\psi(x,t) = \hat{x}\left(\hat{p}_x\psi(x,t)\right) = x\left(-i\hbar\frac{\partial}{\partial x}\psi(x,t)\right) = -i\hbar x\frac{\partial}{\partial x}\psi(x,t). \tag{1.13.3}$$

Thus, we see that $\hat{p}_x\hat{x}\psi(x,t) \neq \hat{x}\hat{p}_x\psi(x,t)$, and

$$\left(\hat{x}\hat{p}_x - \hat{p}_x\hat{x}\right)\psi(x,t) = i\hbar\,\psi(x,t) \tag{1.13.4}$$

Denoting $\hat{x}\hat{p}_x - \hat{p}_x\hat{x}$ by $[\hat{x},\,\hat{p}_x]$, called the commutator between $\hat{x}$ and $\hat{p}_x$,

$$[\hat{x},\,\hat{p}_x] = i\hbar. \tag{1.13.5}$$

Because $[\hat{x},\,\hat{p}_x] \neq 0$, we say $\hat{x}$ and $\hat{p}_x$ do not commute.

On the other hand,

$$[\hat{x},\,\hat{p}_y]\psi(\vec{r},t) \quad = \quad x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r},t) - (-i\hbar\frac{\partial}{\partial y}(x\psi(\vec{r},t)))$$

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$$= \quad x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r},t) + x(i\hbar\frac{\partial}{\partial y}(\psi(\vec{r},t))) = 0, \tag{1.13.6}$$

and $\hat{x}$ and $\hat{p}_y$ commute.

Similarly and collecting,

$$[\hat{x},\,\hat{p}_x] = [\hat{y},\,\hat{p}_y] = [\hat{z},\,\hat{p}_z] = i\hbar \tag{1.13.7}$$

and the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ commute for any other combination, such as

$$[\hat{x},\,\hat{p}_y] = [\hat{y},\,\hat{p}_x] = [\hat{x},\,\hat{y}] = [\hat{p}_x,\,\hat{p}_y] = 0. \tag{1.13.8}$$

We will investigate later that when two operators do not commute there is uncertainty in measuring them simultaneously.

Denoting the components of the vectors by numerical indices with $x = x_1$, $y = x_2$, $z = x_3$, $p_x = p_1$, $p_y = p_2$, and $p_z = p_3$, the commutation relations among the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ can be summarized as

$$[\hat{x}_i,\,\hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i,\,\hat{x}_j] = [\hat{p}_i,\,\hat{p}_j] = 0, \tag{1.13.9}$$

where δ_{ij} is the Kronecker delta defined as $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$.

Exercise 1.13.1: Determine $[\hat{x},\,\hat{p}_x^2]$.

Solution: $[\hat{x},\,\hat{p}_x^2] = \hat{x}\hat{p}_x^2 - \hat{p}_x^2\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{x}\hat{p}_x + \hat{p}_x\hat{x}\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = [\hat{x},\,\hat{p}_x]\hat{p}_x + \hat{p}_x[\hat{x},\,\hat{p}_x] = i\hbar\hat{p}_x + \hat{p}_x i\hbar = 2i\hbar\hat{p}_x$.

In general, given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, we have $[\hat{A},\,\hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A}$, where we have subtracted and added $\hat{B}\hat{A}\hat{C}$. Therefore, $[\hat{A},\,\hat{B}\hat{C}] = [\hat{A},\,\hat{B}]\hat{C} + \hat{B}[\hat{A},\,\hat{C}]$. Thus,

$$[\hat{A},\,\hat{B}\hat{C}] = \hat{B}[\hat{A},\,\hat{C}] + [\hat{A},\,\hat{B}]\hat{C}. \tag{1.13.10}$$

1.14 Time-independent Schrödinger equation

Suppose the potential energy is independent of time; i.e., $V(\vec{r},t) = V(\vec{r})$. The Schrödinger equation then becomes

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(\vec{r})\psi(\vec{r},t), \tag{1.14.1}$$

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Because the potential no longer mixes position and time, we can use method of separation of variables and start with the ansatz

$$\psi(\vec{r},t) = \psi(\vec{r})\phi(t). \tag{1.14.2}$$

With this ansatz, (1.14.1) becomes

$$i\hbar\psi(\vec{r})\frac{\partial\phi(t)}{\partial t} = -\phi(t)\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\phi(t)\psi(\vec{r}). \tag{1.14.3}$$

Dividing both sides by $\psi(\vec{r})\phi(t)$,

$$i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = \frac{1}{\psi(\vec{r})}\left[-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\right]. \tag{1.14.4}$$

Because the left hand side is a function of only t and the right hand side is a function of only $\vec{r}$, each side must be a constant. Let the constant be E. The equation of motion for $\phi(t)$ is then

$$i\hbar\frac{d\phi(t)}{dt} = E\phi(t), \tag{1.14.5}$$

and the solution is

$$\phi(t) = e^{-\frac{i}{\hbar}Et}, \tag{1.14.6}$$

where we have simply set constant coefficient to one, since any constant can be absorbed in $\psi(\vec{r})$.

The equation of motion of $\psi(\vec{r})$ is

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\psi(\vec{r}) = E\psi(\vec{r}), \tag{1.14.7}$$

or

$$\hat{H}\psi(\vec{r}) = E\psi(\vec{r}), \tag{1.14.8}$$

where $\hat{H} = -\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(\vec{r})$ is the Hamiltonian operator. This is called the time-independent Schrödinger equation.

Observe that when the potential is time-independent, the time dependence of the wavefunction that corresponds to the constant E is easily written once we have solved the time-independent Schrödinger equation,

$$\psi_E(\vec{r},t) = \psi_E(\vec{r})e^{-\frac{i}{\hbar}Et}, \tag{1.14.9}$$

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where we have added subscript E to identify $\psi_E(\vec{r},t)$ as a wavefunction that corresponds to constant E.

1.15 Properties of solutions for time-independent potentials

Since we will be investigating only time-independent potentials in nearly all of this course, it is important to summarize some important properties of the solutions when the potential is time-independent.

1.15.1 The constant E must be real

One way that the reality of E follows is from Born's hypothesis on the normalizability of the wavefunction. In particular, considering $\psi_E(\vec{r},t) = \psi_E(\vec{r})\, e^{-\frac{i}{\hbar}Et}$ and writing $E = E_R + iE_I$, where E_R and E_I are possible real and imaginary parts of E, we have $|\psi_E(\vec{r},t)|^2 = |\psi_E(\vec{r})|^2 e^{\frac{i}{\hbar}(E_R - iE_I)t} e^{-\frac{i}{\hbar}(E_R+iE_I)t} = |\psi_E(\vec{r})|^2 e^{\frac{2}{\hbar}E_I t}$, and

$$\int |\psi_E(\vec{r},t)|^2 d^3\vec{r} = \int |\psi_E(\vec{r})|^2 d^3\vec{r} e^{\frac{2}{\hbar}E_I t}, \tag{1.15.1}$$

which cannot be set equal to one for all t except when $E_I = 0$. Therefore, E must be real.

1.15.2 Stationary wavefunction

Given the wavefunction $\psi_E(\vec{r},t)$ that corresponds to E,

$$\mathcal{P}(\vec{r},t) = |\psi_E(\vec{r},t)|^2 = |\psi_E(\vec{r})|^2, \tag{1.15.2}$$

and the probability density is independent of time. We say the wavefunction $\psi_E(\vec{r},t)$ is stationary.

1.15.3 Stationary wavefunction is a state of definite energy

Using $\hat{H}\psi_E(\vec{r},t) = E\psi_E(\vec{r},t)$, the expectation value of the Hamiltonian operator in state $\psi_E(\vec{r})$ is

$$\langle \hat{H} \rangle = \int \psi_E(\vec{r},t)^* \hat{H} \psi_E(\vec{r},t) d^3\hat{r} = \int \psi_E(\vec{r},t)^* E \psi_E(\vec{r},t) d^3\hat{r} = E \int |\psi_E(\vec{r},t)^2 d^3\hat{r} = E. \tag{1.15.3}$$

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Therefore, the constant E is the expectation value of the Hamiltonian operator, and it is energy. Furthermore,

$$\langle \hat{H}^2 \rangle = \int \psi_E(\vec{r},t)^* \hat{H}^2 \psi_E(\vec{r},t) d^3\hat{r} = \int \psi_E(\vec{r},t)^* EE\psi_E(\vec{r},t) d^3\hat{r} = E^2 \int |\psi_E(\vec{r},t)^2 d^3\hat{r} = E^2, \tag{1.15.4}$$

and the uncertainty or variance of the energy is

$$(\Delta E) = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} = \sqrt{E^2 - E^2} = 0. \tag{1.15.5}$$

Therefore, the stationary state $\psi_E(\vec{r},t)$ is a state of definite energy E.

1.15.4 Spatial part of stationary wavefunction can be chosen to be real

Given $\psi_E(\vec{r})$ is a solution to the time-independent Schrodinger equation with energy E

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \tag{1.15.6}$$

and complex conjugating it, we have

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r})^* + V(\vec{r})\psi_E(\vec{r})^* = E\psi_E(\vec{r})^*. \tag{1.15.7}$$

In other words, $\psi_E(\vec{r})^*$ is also a solution with the same energy. Therefore, we can trade $\psi_E(\vec{r})$ and $\psi_E(\vec{r})^*$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(\vec{r})^* \qquad \text{and} \qquad \frac{\psi_E(\vec{r}) - \psi_E(\vec{r})^*}{i}, \tag{1.15.8}$$

which are both real. Thus, the spatial part of a stationary wavefunction can be chosen to be real.

1.15.5 Parity symmetry: Even and odd wavefunctions

Suppose the potential is symmetric under reflection about the origin, $V(-\vec{r}) = V(\vec{r})$. In this case, given

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \tag{1.15.9}$$

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and replacing $\vec{r}$ by $-\vec{r}$ and setting $V(-\vec{r}) = V(\vec{r})$, we have a

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(-\vec{r}) + V(\vec{r})\psi_E(-\vec{r}) = E\psi_E(-\vec{r}). \tag{1.15.10}$$

Therefore, $\psi_E(-\vec{r})$ is a solution with the same energy. We can trade $\psi_E(\vec{r})$ and $\psi_E(-\vec{r})$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(-\vec{r}) \qquad \text{and} \qquad \psi_E(\vec{r}) - \psi_E(-\vec{r}). \tag{1.15.11}$$

Now $\psi_E(\vec{r}) + \psi_E(-\vec{r})$ is even under reflection while $\psi_E(\vec{r}) - \psi_E(-\vec{r})$ is odd. Thus, when the potential is symmetric under reflection, we can choose the stationary states to be either even or odd under reflection.

1.15.6 Orthogonality/orthonormality

Suppose $\psi_n(\vec{r}, t)$ is a solution with energy E_n and $\psi_m(\vec{r}, t)$ is a solution with energy E_m; i.e.,

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_n(\vec{r}) = E_n\psi_n(\vec{r}) \quad \text{and} \quad -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_m(\vec{r})^* = E_m\psi_m(\vec{r})^*, \tag{1.15.12}$$

where we have taken the complex conjugate for the second. Multiplying both sides of the first by $\psi_m(\vec{r})^*$ and the second by $\psi_n(\vec{r})$,

$$\begin{aligned} -\frac{\hbar^2}{2m}\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_m(\vec{r})^*\psi_n(\vec{r}) &= E_n\psi_m(\vec{r})^*\psi_n(\vec{r}) \\ -\frac{\hbar^2}{2m}\psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_n(\vec{r})\psi_m(\vec{r})^* &= E_m\psi_n(\vec{r})\psi_m(\vec{r})^* \end{aligned} \tag{1.15.13}$$

Subtracting the two,

$$-\frac{\hbar^2}{2m}\left(\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) - \psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^*\right) = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}) \tag{1.15.14}$$

Recalling our computation and definition of the probability current, the left hand side equals $-i\hbar\vec{\nabla}\cdot\vec{J}$. Therefore,

$$-i\hbar\vec{\nabla}\cdot\vec{J} = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}). \tag{1.15.15}$$

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Integrating both sides, and using the divergence theorem, $\int \vec{\nabla} \cdot \vec{J} d^3\vec{r} = \oint \vec{J} \cdot d\vec{a}$,

$$-i\hbar \oint \vec{J} \cdot d\vec{a} \quad = \quad (E_n - E_m) \int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r}. \tag{1.15.16}$$

Now, taking the boundary of the area integral to infinity, the current vanishes, since a normalizable wavefunction and the corresponding current must vanish at infinity, and the left hand side vanishes. We then have

$$(E_n - E_m) \int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 0. \tag{1.15.17}$$

We have two cases.

1. Case 1: $n = m$ and $E_n = E_m$. In this case, $E_n - E_m = 0$ and for a normalized wavefunction $\psi_n(\vec{r})$, $\int \psi_n(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 1$.
2. Case 2: $n \neq m$ and $E_n \neq E_m$. In this case, $E_n - E_m \neq 0$ and $\int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 0$. We say $\psi_n(\vec{r})$ and $\psi_m(\vec{r})$ are orthogonal.

Combining the two cases, we have the orthonormality relation

$$\int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = \delta_{mn}. \tag{1.15.18}$$

1.15.7 Linearity

The Schrödinger equation is linear. Given stationary states $\psi_n(\vec{r}, t) = \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, $n = 1, 2, ...$, a linear combination of these

$$\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}, \tag{1.15.19}$$

where c_n are complex constants, is a solution to the time-dependent Schrödinger equation $i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = \hat{H}\psi(\vec{r}, t)$. To check that, the left hand side is $i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = \sum_n c_n \psi_n(\vec{r}) i\hbar \frac{\partial}{\partial t} e^{-\frac{i}{\hbar} E_n t} = \sum_n c_n E_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, and the right hand side is $\hat{H}\psi(\vec{r}, t) = \sum_n c_n \hat{H}\psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t} = \sum_n c_n E_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, and they are equal.

1.15.8 Time evolution

Because $\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$ with constant coefficients c_n is a solution to the time-dependent Schrödinger equation, and $\psi(\vec{r}, 0) = \sum_n c_n \psi_n(\vec{r}, 0) = \sum_n c_n \psi_n(\vec{r})$ at time

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$t = 0$, we note that given

$$\psi(\vec{r}, 0) = \sum_n c_n \psi_n(\vec{r}, 0) = \sum_n c_n \psi_n(\vec{r}) \tag{1.15.20}$$

at time $t = 0$, the time evolution, or the wavefunction at time t, is simply

$$\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}. \tag{1.15.21}$$

In other words, once we have expanded a given initial wavefunction in terms of a linear combination of the stationary wavefunctions $\psi_n(\vec{r})$, the time evolution follows simply by putting a factor of $e^{-\frac{i}{\hbar} E_n t}$ to each term containing $\psi_n(\vec{r})$.

1.15.9 Normalization

Given $\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$ with complex coefficients c_n, normalizing it leads to

$$\begin{aligned} 1 &= \int |\psi(\vec{r}, t)|^2 d^3\vec{r} = \int \left(\sum_m c_m^* \psi_m^*(\vec{r}) e^{\frac{i}{\hbar} E_m t} \right) \left(\sum_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t} \right) d^3\vec{r} \\ &= \sum_n \sum_m c_m^* c_n e^{\frac{i}{\hbar}(E_m - E_n)t} \int \psi_m^*(\vec{r}) \psi_n(\vec{r}) d^3\vec{r} = \sum_n \sum_m c_m^* c_n e^{\frac{i}{\hbar}(E_m - E_n)t} \delta_{mn} = \sum_n |c_n|^2. \end{aligned} \tag{1.15.22}$$

Therefore, the constant coefficients are constrained by

$$\sum_n |c_n|^2 = 1. \tag{1.15.23}$$

1.15.10 Completeness

Given an arbitrary $\psi(\vec{r}, t)$ that is a solution to the Schrödinger equation with certain boundary conditions, we can express it as a linear superposition of the stationary solutions $\psi_n(\vec{r}, t) = \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$ as

$$\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}. \tag{1.15.24}$$

with complex coefficients c_n. In other words, the stationary orthonormal solutions form a complete set.

To find the condition for the above statement to hold, first let us find the coefficients c_n. Mul-

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tiplying both sides by $\psi_m(\vec{r},t)^*$ and integrating over $d^3\vec{r}$,

$$\int \psi_m(\vec{r},t)^*\psi(\vec{r},t)d^3\vec{r} = \sum_n c_n \int \psi_m(\vec{r},t)^*\psi_n(\vec{r},t)d^3\vec{r} = \sum_n c_n\delta_{mn} = c_m. \tag{1.15.25}$$

Therefore,

$$c_n = \int \psi_n(\vec{r},t)^*\psi(\vec{r},t)d^3\vec{r}. \tag{1.15.26}$$

Inserting these coefficients back into the expansion,

$$\begin{aligned}\psi(\vec{r},t) &= \sum_n \int \psi_n(\vec{r}',t)^*\psi(\vec{r}',t)d^3\vec{r}'\psi_n(\vec{r},t) \\ &= \int d^3\vec{r}'\psi(\vec{r}',t)\sum_n \psi_n(\vec{r}',t)^*\psi_n(\vec{r},t).\end{aligned} \tag{1.15.27}$$

The stationary states form a complete set if

$$\sum_n \psi_n(\vec{r}',t)^*\psi_n(\vec{r},t) = \delta^3(\vec{r}'-\vec{r}), \tag{1.15.28}$$

where $\delta^3(\vec{r}'-\vec{r})$ is the Dirac-delta function in three dimensions defined by

$$\int d^3\vec{r}'\psi(\vec{r}',t)\delta^3(\vec{r}'-\vec{r}) = \psi(\vec{r}',t). \tag{1.15.29}$$

In other words, if we multiply and sum the stationary states as $\sum_n \psi_n(\vec{r}',t)^*\psi_n(\vec{r},t)$ and we get $\delta^3(\vec{r}'-\vec{r})$, we know we have a complete set. Conversely, if we miss a state in the sum, we do not get $\delta^3(\vec{r}'-\vec{r})$. In cartesian coordinates, $\delta^3(\vec{r}'-\vec{r}) = \delta(x-x')\delta(y-y')\delta(z-z')$.

1.16 Delta function

Below, we summarize some of the properties of the delta function, focusing on one dimension. Given a function $f(x)$, the delta function $\delta(x-x')$ is defined such that

$$f(x') = \int f(x)\delta(x-x')dx \tag{1.16.1}$$

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In other words, $\delta(x-x')$ in the integral projects out the value of $f(x)$ at the zero of the argument $x-x'$ of $\delta(x-x')$. Therefore, we can view $\delta(x-x')$ as defined by

$$\delta(x-x')=0 \text{ for } x\neq x', \qquad f(x')=\int_a^b f(x)\delta(x-x')dx \text{ for } a<x' \text{ and } b>x'. \tag{1.16.2}$$

Because $\delta(x-x')=0$ for $x\neq x'$, we can rewrite $f(x')=\int f(x)\delta(x-x')dx=f(x')\int\delta(x-x')dx$. Therefore,

$$\int \delta(x-x')dx=1. \tag{1.16.3}$$

However, the delta function is not actually a function but a distribution. For instance, we can not interpret $\int\delta(x-x')dx=1$ as the area under $\delta(x-x')$ being one.

Here we list some representations of the delta function,

$$\delta_\alpha(x) = \frac{1}{\alpha\sqrt{\pi}}e^{-\frac{x^2}{a^2}} \tag{1.16.4}$$

$$\delta_\alpha(x) = \frac{1}{\pi x}\sin\left(\frac{x}{\alpha}\right) \tag{1.16.5}$$

$$\delta_\alpha(x) = \frac{\alpha}{\pi x^2}\sin^2\left(\frac{x}{\alpha}\right) \tag{1.16.6}$$

With any of these representations, the delta function $\delta_\alpha(x-a)$ is defined through

$$f(x')=\lim_{\alpha\to 0}\int f(x)\delta_\alpha(x-x')dx. \tag{1.16.7}$$

However, $\lim_{\alpha\to0}\int f(x)\delta_\alpha(x-x')dx\neq\int\lim_{\alpha\to0}f(x)\delta_\alpha(x-x')dx$. Indeed the integral vanishes if the limit is taken within the integral.

Exercise 1.16.1: Show the following.

1. $\delta(ax)=\frac{1}{|a|}\delta(x)$
2. $\delta(x^2-a^2)=\frac{1}{2|a|}\left(\delta(x+a)+\delta(x-a)\right)$
3. $\delta(f(x))=\sum_i\frac{\delta(x-x_i)}{|f'(x)|}$, where x_i are the roots of $f(x)=0$.
4. Write $\delta^3(\vec{r}-\vec{r}')$ in spherical coordinates.
5. Write $\delta^3(\vec{r}-\vec{r}')$ in cylindrical coordinates.

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